

Harshal Bhambhani

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My Portfolio

Professional Summary

AI Engineer with production experience building Generative AI systems in BFSI at OTO Capital and Standard Chartered GBS. Built multi-tenant AI voice-collections stack with Gemini Live signal extraction serving 8 API endpoints, and production NLP pipelines for financial document compliance. Hands-on with LLMs, prompt engineering, LLM evaluation, and Python API development for lending and collections domains.

Skills

Generative AI & LLMs: LLMs, Prompt Engineering, LLM Evaluation, RAG, Chatbots, Gemini, LangChain, FAISS, Fine-Tuning

ML & Data Science: Machine Learning, Scikit-Learn, XGBoost, SHAP, NLP, Sentiment Analysis, Statistics

BFSI & Analytics: Python, SQL, PostgreSQL, FastAPI, REST API, Streamlit, Power BI, Pandas, NumPy

DevOps & Tools: Odoo 19, Git, GitHub, API Integration, Exotel, Docker

Cloud & MLOps: AWS S3, Docker, Kubernetes, GitHub

Work Experience

AI Engineer - OTO Capital Aug 2026 – Present

- Architected multi-tenant AI voice-collections stack in Python and FastAPI, shipping 8 API endpoints.
- Engineered Gemini Live signal-extraction layer emitting 7 disposition tags, widening call coverage 7x.
- Diagnosed 1.59M collection calls in PostgreSQL, exposing a 1,300x cost-per-payment gap in cohorts.

Data & Analytics Consultant - Goel Exports July 2026 – Aug 2026

- Reverse-engineered a furniture exporter's order-to-cash process into a documented 4-phase Odoo 19 data model.
- Architected a critical-datapoint framework across 6 pipeline stages, driving field-level mandatory-config rules.
- Diagnosed CRM automation defects via Odoo system logs, resolving tag-trigger and activity-routing misassignments.

Data Science Intern - Standard Chartered Global Business Services Pvt. Ltd. Jul 2025 – Dec 2025

- Built production Python NLP pipeline parsing 100+ page loan contracts, cutting manual compliance audit time by 90%.
- Engineered high-recall text classifiers achieving 98.6% recall on rare clauses, mitigating client risk at scale.
- Configured SBERT vector search with cosine similarity for gold-standard clause retrieval across loan documents.
- Embedded confidence-threshold guardrails filtering 95%+ false positives, delivering audit-grade analytical outputs.
- Streamlined ETL pipeline using 250-token sliding windows and 50-token overlaps, achieving zero data loss on files.

Machine Learning Engineer Intern - DG Liger Consulting Jun 2024 – Jul 2024

- Analysed 30+ proprietary PDFs using LangChain document loaders, structuring client data into queryable pipelines.
- Indexed semantic embeddings into FAISS vector stores enabling sub-second similarity search across document sets.
- Integrated 3B-parameter LLMs grounding retrieval-augmented answers in domain-specific business data for team leads.

Technical Projects

Market Analysis Using LLMs — Trend Analysis & Statistical Rebalancing Jun 2024 – July 2024

- Orchestrated multi-stage NLP pipeline: BART summarisation and RoBERTa sentiment classification on 500+ articles.
- Deployed real-time Flask REST API serving ML predictions with Streamlit monitoring dashboard for trend visualisation.
- Curated 500+ financial articles via BeautifulSoup web scraping scripts, applied SMOTE rebalancing for classifiers.

Multilingual ASR System - Signal Processing & Measurement Pipeline Aug 2024 – Dec 2024

- Transcribed low-resource Urdu speech via Whisper zero-shot with language biasing tokens, reducing Word Error Rate
- Corrected ASR output errors via IndicBERT masked language model, replacing low-confidence tokens at 0.4 threshold.
- Normalised raw audio via pydub pipeline: mono conversion, -20dBFS loudness, silence trimming, and 16kHz resampling.

Hackathon / Datathon

Medical Bill OCR & Fraud Detection - Bajaj Finserv Health Nov 2025

- Processed scanned hospital bills via PaddleOCR with CLAHE enhancement and 4x DPI scaling for 90%+ accuracy rates.
- Parsed raw OCR text into structured JSON via Qwen 7B LLM with schema-constrained prompts across bill templates.
- Automated 4 parallel anomaly detection algorithms using IQR statistical analysis and SHA-256 fraud hashing.

Education

B.E. Electrical and Electronics with a Minor in Computer Science BITS Pilani, Hyderabad(2022-2026)
CBSE: Class 12 - 92.2% — Class 10 - 91.2%